UNIT 1: Choice of architectures and programming tools.

Exercises IV

1. What does DevOps combine?

   A. Design and Programming
   B. Development and Operations
   C. Development and Testing
   D. Deployment and Security

2. Which of the following best describes DevOps?

   A. A programming language
   B. A database technology
   C. A culture and set of practices that encourage collaboration and automation
   D. A tool used only for deploying applications

3. What does CI stand for?

   A. Continuous Integration
   B. Continuous Installation
   C. Code Integration
   D. Continuous Inspection

4. What is one of the main purposes of Continuous Integration?

   A. To avoid testing software
   B. To integrate code frequently and detect problems early
   C. To replace developers
   D. To manually deploy applications

5. What does CD commonly refer to?

   A. Code Development
   B. Continuous Delivery / Continuous Deployment
   C. Central Database
   D. Code Distribution

6. DevOps means that developers and operations teams should work together (true/false). **true**
7. CI means that developers should wait several weeks before integrating their code(true/false). **false**
8. Automated tests can help detect problems shortly after code changes are made (true/false). **true**
9. Continuous Delivery means that software is always kept in a state where it can be reliably released (true/false).

   true

10. Continuous Deployment always requires a person to manually approve every production release (true/false).

    false

11. Automation is an important part of DevOps (true/false). **true**

12. In your own words, explain what DevOps means.

    Es una cultura de practicas para que los diferentes grupos puedan trabajar juntos y entregar el software antes

13. Why is it useful for developers and operations teams to collaborate?

    Porque construyen el software antes al repartirse el trabajo entre los diferentes equipos y despues ponerse en comun y acaban con un resultado de mejor calidad

14. Explain what happens when a developer pushes new code to a repository in a typical CI process.

    Primero se desarrolla la aplicación luego se mira que pasen los test y si estan bien se continuan si no se arregla el codigo ou se avisa ao equipo de desenvolvemento

15. Why are automated tests important in CI?

    Porque asi ayuda a descubrir los problemas antes y al estar automatizados puedes centrarte mas en desarrollar la aplicación y los errores se corregirian antes

16. What is the difference between Continuous Delivery and Continuous Deployment?

    La diferenciae ntre el continuous delivery y el continuos deployment es que en el delivery cuando se pasan los test aun hay una persona que tiene que aprobar que se lance la aplicación mientra que en el continuous deployment una vez pasan los test ya se publica la aplicación automaticamente

17. For each situation, decide whether it is mainly related to DevOps, CI, Continuous Delivery, or Continuous Deployment.

    A) A team automatically runs unit tests every time a developer pushes code. **ci**

    B) Developers and system administrators work together instead of working as completely separate teams. **devOps**

    C) After passing all tests, an application is automatically deployed to the production server.

       Continous deployment

    D) An application is automatically built, tested, and packaged so that it is ready to be released, but a manager must approve the production release.

       Continous Delivery

18. Situation analysis:

Five developers are working on the same application.

Developer A changes the login system. Developer B changes the database code. Developer C changes the user interface.

They don’t integrate their changes until the end of the month.

When they finally combine everything, the application doesn’t work.

Questions:

- What problems can you identify?

  Estan construyendo la aplicación totalmente por separado y no van a poder ir arreglando los fallos o problemas de compatibilidad hasta que lo junten todo
- How could CI help this team?

  Podran encontrar el error antes porque ya se juntara todo y podran ver si pasan los test o no
- What would happen if the team integrated and tested their changes frequently?

  Podran encontrar los errores antes y poder arreglarlos antes

19. Situation analysis:

A company deploys its application manually.

Every Friday, an administrator:

1. Copies files to the server.
2. Changes configuration files.
3. Runs some tests manually.
4. Restarts the application.

Sometimes the administrator forgets a step.

Questions:

1. What risks do you see in this process?

Puede ser que se pierdan algunos archivos copiandolos o puede haber problemas con los cambios de configuracion inluso pueden faltar algunos de los test por comprobar por despiste de no pasarlos

2. How could automation help?

Te evitarias todo eso ya que se subiria a un repositorio conjunto y ya pasaria los test

3. How could CD improve the deployment process?

Podria publicarse la aplicación con menos fallos porque no deberia haber errores humanos y usando continous delivery podria un humano revisar que todo esta bien antes

de publicarse para asegurarse

20. Designing a Pipeline

Imagine that you are responsible for a web application.

Design a simple CI/CD pipeline.

Include at least five steps.

For example:

Code
↓
build
↓
test
↓
human approval
↓
Production

Explain what happens at each step.

Primero escribes el codigo luego ese codigo se pasa a un repositorio comun para construir la aplicación pasa los test una vez pasan los test si son correctos un humano revisa que este todo bien y ya se lanza

21. Production Bug

A developer pushes new code. The CI pipeline runs, but one automated test fails.

The developer says:

“Let’s deploy it anyway. It’s probably not important.”

Would you deploy the application?

No primero habria que revisar que error esta dando y intentar desplegar la aplicación sin errores

Explain your answer.